

This assignment will not be graded and is only for practice.

1) Choose the set of correct options.

1 point

- ☐ Any subset of any linearly independent set of vectors is linearly independent.
- ☐ Any subset of any linearly dependent set of vectors is linearly dependent.
- ☐ Any subset of any linearly dependent set of vectors is linearly independent.
- ☐ Any set of 3 vectors in $\mathbb{R}^3$ is linearly independent.
- ☐ Any set of 4 vectors in $\mathbb{R}^3$ is linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Any subset of any linearly independent set of vectors is linearly independent.

Any set of 4 vectors in $\mathbb{R}^3$ is linearly dependent.

Recall the definition of linearly dependent and independent:

Statement 1) Definition of linear dependence: A set of vectors $\{v_1, v_2, \dots, v_n\}$ in V is said to be linearly dependent if there exist scalars $a_1, a_2, \dots, a_n \in \mathbb{R}$, not all zero such that $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$.

Statement 2) Definition of linear independence: A set of vectors $\{v_1, v_2, \dots, v_n\}$ in V is said to be linearly independent if $a_1v_1 + a_2v_2 + \dots + a_nv_n = 0$ implies that $a_i = 0$ for $i = 1, 2, \dots, n$.

Statement 3) If S is a set containing zero vector, then the set S is a linearly dependent set.

Statement 4) Let S be a subset of a vector space V . If an element of S is a scalar multiple of another element from the set S , then S is a linearly dependent set.

Statement 5) If S is linearly dependent, then there exists $v \in S$, such that v can be written as a linear combination of other vectors in S and vice versa.

Discussion on Option 1: Let $S = \{v_1, v_2, \dots, v_n\}$ be a linearly independent set of vectors and let S' be a subset of S .

Claim: S' is linearly independent.

2) What we have to assume for S' if we want to use the method of contradiction?

1 point

- ☐ Let us assume that S' is linearly dependent.

☐ Let us assume that S' is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let us assume that S' is linearly dependent.

If S' is linearly dependent, then there exists $v \in S'$, such that v can be written as a linear combination of other vectors in S' .

3) Which of the above statements we are using here?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

$v \in S'$ implies $v \in S$, as $S' \subseteq S$. Those vectors of whose linear combination is giving v are also in S due to the same argument.

4) Again from statement 5, can we conclude that S is linearly dependent?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

But S was given to be a linearly independent set, hence we arrive at a contradiction. So our claim is proved.

Another method: Let $S' = \{v_{i_1}, v_{i_2}, \dots, v_{i_k}\} \subseteq S$. Consider the equation

$$a_1 v_{i_1} + a_2 v_{i_2} + \dots + a_k v_{i_k} = 0$$

5) Can we conclude each $a_i = 0$ from here?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

As all the v_{i_j} are also in S , the given equation is a linear combination of vectors from S . Now as S is linear independent, $a_i = 0$ for each $i \in \{1, 2, \dots, k\}$. Hence S' is linearly independent.

· Now try Question 7 and 8 of activity 3.3

Discussion on Option 2: Let us discuss an example. Let $S = \{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$ be a subset of $\mathbb{R}^3$.

6) Is S linearly dependent?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: $(1, 1, 0) = (1, 0, 0) + (0, 1, 0)$

Accepted Answers:

Yes.

7) Can you find a linearly independent subset of S ?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Check that any proper subset of S of cardinality 1 or 2 is linearly independent subset of S .

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (0, 1, 0)\}$.

8) Is $S' \subseteq S$?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

9) Is S' linearly independent?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Hence Option 2 is not correct.

Discussion on Option 3: Here also we construct an example. Let $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$.

10) Is S linearly dependent?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

Consider $S' = \{(1, 0, 0), (2, 0, 0)\}$.

11) Is $S' \subseteq S$?

1 point

☐ Yes.

☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

12) Is S' linearly independent?**1 point**☐ Yes.☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 3 is not correct.

Discussion on Option 4: Consider the same set $S = \{(1, 0, 0), (2, 0, 0), (3, 0, 0)\}$ that we have discussed previously.

13) Is S linearly independent?**1 point**☐ Yes.☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

14) What can we conclude about Option 4 from this?

1 point

- ☐ Option 4 is correct.
- ☐ Option 4 is not correct.
- ☐ Nothing can be concluded about Option 4 from here.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Option 4 is not correct.

· Now try Question 5 and 8 of activity 3.4

Discussion on Option 5: Let $S = \{v_1, v_2, v_3, v_4\}$ be a set of 4 vectors in $\mathbb{R}^3$. Let $v_1 = (x_1, y_1, z_1)$, $v_2 = (x_2, y_2, z_2)$, $v_3 = (x_3, y_3, z_3)$, $v_4 = (x_4, y_4, z_4)$. Consider the following equation,

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$$

i.e.,

$$a_1(x_1, y_1, z_1) + a_2(x_2, y_2, z_2) + a_3(x_3, y_3, z_3) + a_4(x_4, y_4, z_4) = (0, 0, 0)$$

i.e.,

$$a_1 x_1 + a_2 x_2 + a_3 x_3 + a_4 x_4 = 0$$

$$a_1 y_1 + a_2 y_2 + a_3 y_3 + a_4 y_4 = 0$$

$$a_1 z_1 + a_2 z_2 + a_3 z_3 + a_4 z_4 = 0$$

The matrix representation of this system of linear equations is given by

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ z_1 & z_2 & z_3 & z_4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We want to find the solution for the vector $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$ for this given homogeneous system of linear equations.

15) What can we say about the solutions of this system of linear equations?

1 point

- ☐ The solution is unique.
- ☐ There are infinitely many solutions.
- ☐ There must exist a non-zero solution.
- ☐ 0 is the only solution.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Try to compute the maximum number of dependent variables you can get from the row reduced echelon form of the above matrix.

Accepted Answers:

There are infinitely many solutions.
There must exist a non-zero solution.

16) Is S linearly independent?

1 point

- ☐ Yes.
- ☐ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 5 is correct.

This will help you to solve Question number 4 of the graded assignment.

Your score is: 0/16